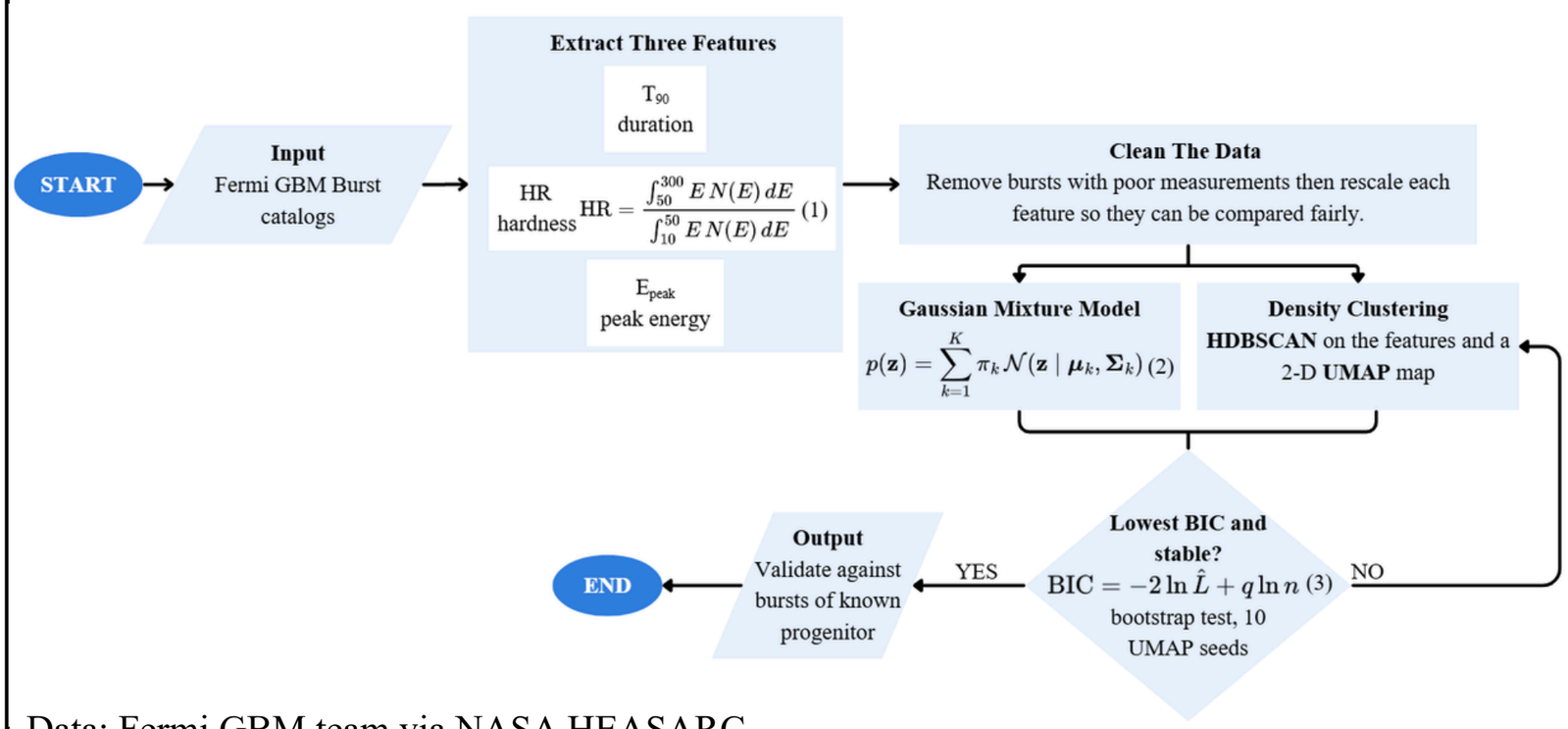
DATA-DRIVEN CLASSIFICATION OF GAMMA-RAY BURSTS USING DURATION, HARDNESS AND SPECTRAL PEAK ENERGY

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INTRODUCTION

Gamma-ray bursts (GRBs) are the **most luminous explosions** in the Universe. Their durations are **bimodal**, split at $T_{90} \approx 2$ s [1]: short GRBs are linked to **compact-object mergers** and long GRBs to **collapsing massive stars**. Recent bursts break this rule: **GRB 211211A** was long but produced a kilonova [3] while **GRB 200826A** was short but produced a supernova [2]. Machine learning on prompt-emission properties finds two main classes [4] but a possible third class is still debated. This study classifies Fermi GBM bursts **without labels** using duration, hardness and peak energy, and tests the clusters against bursts with **known progenitors**.

METHODOLOGY



Data: Fermi GBM team via NASA HEASARC.
Software: Astropy, scikit-learn, hdbscan, umap-learn

RESULT

- Durations are **bimodal**: short bursts (median 0.8 s, 20%) and long bursts (26 s, 80%).
- BIC prefers **two classes** for duration alone ($\Delta BIC = +14.5$ for three).
- A bootstrap test finds extra structure ($p < 0.005$) but **no clear third class**.
- UMAP + HDBSCAN finds the same **two clusters** in all 10 random seeds.

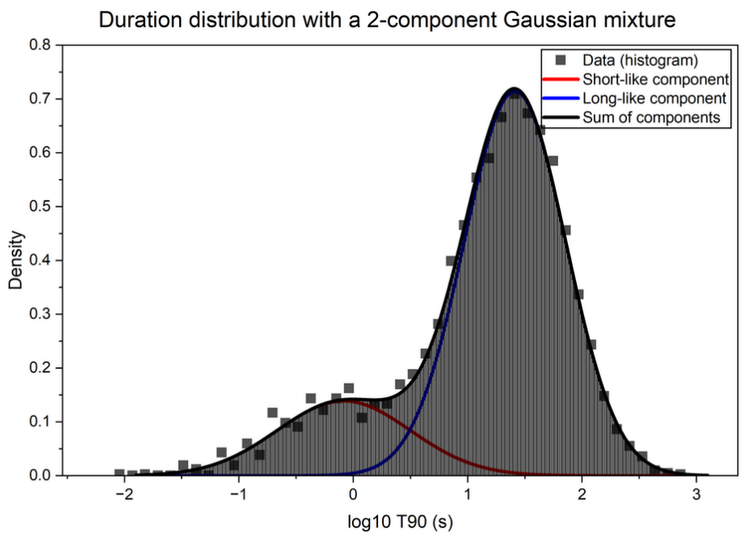


Figure 1 log T_{90} with a 2-class Gaussian mixture.

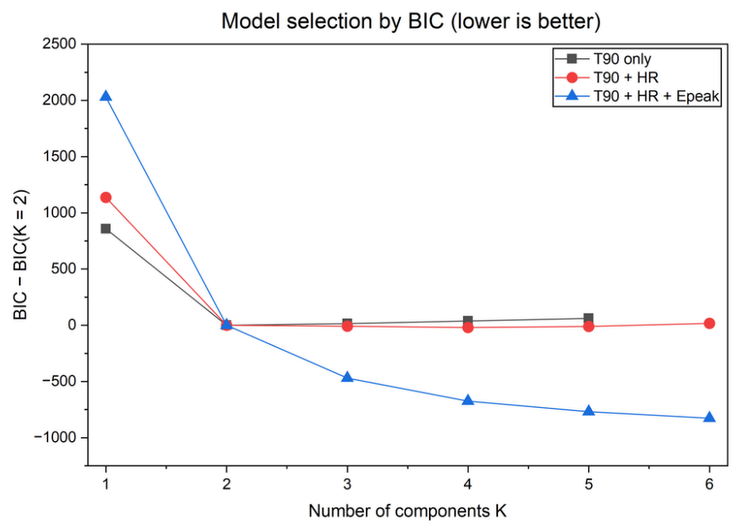


Figure 2 BIC relative to K=2.

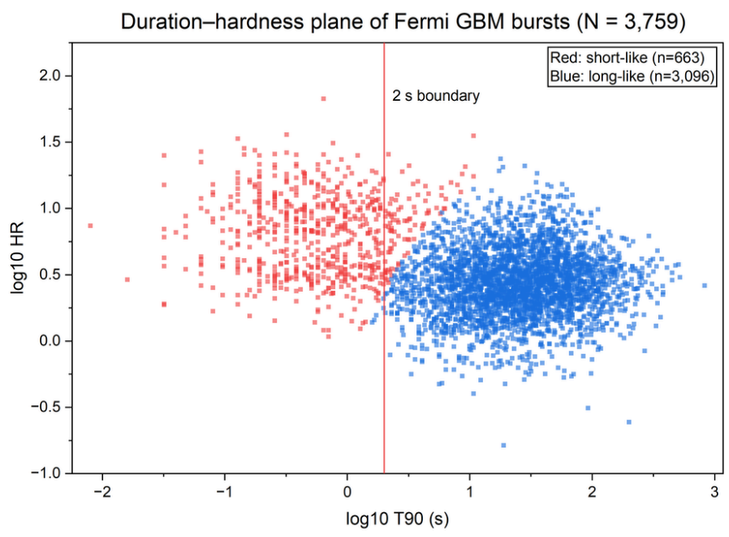


Figure 3 2-class GMM matches the 2 s cut for 97.5% of bursts; short bursts are ~2.5× harder.

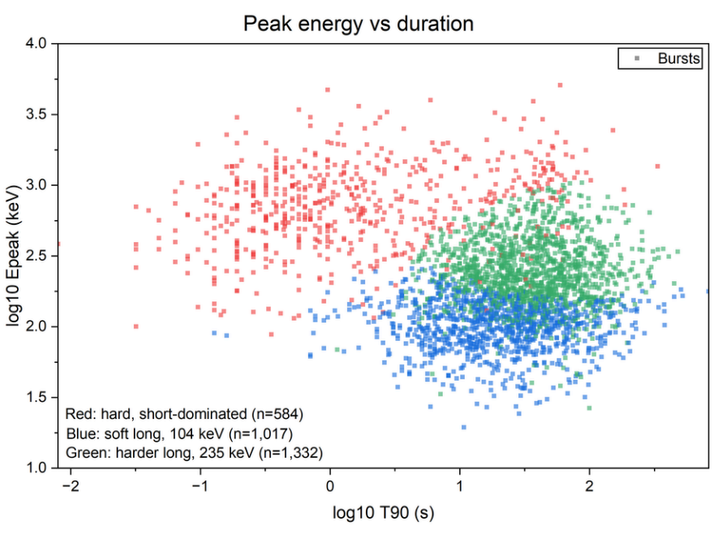


Figure 4 E_{peak} splits long bursts into soft (104 keV) and hard (235 keV) groups, not an intermediate class.

Gamma-Ray Burst (GRB)	Origin	Gaussian Mixture Model (GMM)	Uniform Manifold Approximation and Projection (UMAP)
090510	merger	short	short
170817A	merger	short	-
200826A	collapsar	short	long
211211A	merger	long	long
130427A	collapsar	long	long

Table 1 GMM uses (T_{90}, HR); UMAP + HDBSCAN also uses E_{peak}. Adding E_{peak} fixes the collapsar GRB 200826A but the merger-origin long GRB 211211A still looks like a normal long burst. Green cells match the known progenitor; red cells do not.

CONCLUSION

Unsupervised learning on Fermi GBM bursts recovers **two dominant GRB classes**, matching the 2 s boundary for 97.5% of bursts with **no robust evidence for a third class.** The extra structure lies within long bursts which split by spectral hardness. Adding the peak energy correctly reclassifies the collapsar GRB 200826A but **merger-origin long bursts** such as GRB 211211A need information beyond these features.

REFERENCES

[1] C. Kouveliotou et al., "Identification of two classes of gamma-ray bursts," Astrophys. J., vol. 413, pp. L101–L104, Aug. 1993, doi: 10.1086/186969.
[2] T. Ahumada et al., "Discovery and confirmation of the shortest gamma-ray burst from a collapsar," Nat. Astron., vol. 5, no. 9, pp. 917–927, Sep. 2021, doi: 10.1038/s41550-021-01428-7.
[3] J. C. Rastinejad et al., "A kilonova following a long-duration gamma-ray burst at 350 Mpc," Nature, vol. 612, pp. 223–227, Dec. 2022, doi: 10.1038/s41586-022-05390-w.
[4] C. L. Steinhardt, W. J. Mann, V. Rusakov, and C. K. Jespersen, "Classification of BATSE, Swift, and Fermi gamma-ray bursts from prompt emission alone," Astrophys. J., vol. 945, no. 1, Art. no. 67, Mar. 2023, doi: 10.3847/1538-4357/acb999.
[5] S. Poolakkil et al., "The Fermi-GBM gamma-ray burst spectral catalog: 10 yr of data," Astrophys. J., vol. 913, no. 1, Art. no. 60, May 2021, doi: 10.3847/1538-4357/abf24d.